Peer review: Theorem I and Theorem $\mathbf{I_{OG}}$

the integer zero window, exactly, on LG$(m, 2m)$ and OG$(m, 2m{+}1)$

endorsed; with a shorter proof of its load-bearing lemma, and a correction to my own closed form

Clio (clio-vega)

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PROTOCOL §2.3 record.

Author	Clio
Date	2026-10-24
Recipient	Rick (grandparick20@gmail.com), cc Robin
Repository	`clio-vega/rick-review`
Commit	`297d85a1f93d4a31fa9fa22dcd47c1e6a5573275`

This PDF corresponds to the pushed commit above. Where this text and the commit disagree, **the commit is authoritative**.

Target	`proofs/2026-10-24-integer-window-and-OG-twin.md` (308 lines)
Reviewed at	`rick-research` `7ff2361d182d258dbf6c829c4aa97bcfd9fffc8a`
Arming stamp	`7bd9ae3c`; `origin/main` had moved to `b44a146` (11 commits)
Covering mail	UID 847; context UIDs 848, 850, 851
Verification code	`clio-vega/rick-review` `297d85a`, dir `code-1024c2-thmI/`

0. First instrument: the commit range

Eleven commits landed between the arming stamp and this session. The target file did *not* move, so I reviewed the version the brief named. Three things in the range matter: PROVE/Dream 254 upgraded many registry nodes `computed`→`proved` (including **Theorem B**, which I had left unendorsed, §6); `0e4001a` gives a BR-free second proof of $\mathrm{SR}_{\mathrm{LG}}$; and the new file `proofs/2026-10-25-crownF-strict-reciprocity.md` contains **Theorem Q30**, which *retires the reader-left dependency Theorem I still declares for itself.* That is Finding 2, and it moves the grade upward.

Inbox re-checked immediately before sending, not at arming: UIDs 849, 850, 851 had landed past the watermark and are answered in §8.

1. The claim, in one sentence

With $s_j(\kappa) := (-1)^{|\kappa|} H_\kappa(-j)$, one monotone induction on $|\kappa|$ — driven by the K-theoretic recursion

$$s_{j+1}(\kappa) \;=\; s_j(\kappa) \;+\; \sum_{\rho} |c_\rho|\, s_j(\rho) \qquad (\star)$$

whose increments are all ≥ 0 *because the anchor* $s_1 = 0$ *leaves the alternating signs nothing to fight* — yields the exact integer zero set $\{-1, \ldots, -W\}$ with $W = \lceil r(\kappa)/2 \rceil$ on LG and $W = r(\kappa)$ on OG, plus the sign $(-1)^{|\kappa|}$ beyond the window.

The mechanism is the best idea in the file, and §11 names it correctly: when an alternating-sign recursion refuses to be coherent, look for the specialisation where the base case vanishes.

2. Verdict

Item	Grade I endorse	Basis
Lemma 0 (chain formula)	proved	510 κ, $m \le 8$, three definitions agree, 0 mismatches
Lemma 1 (anchor $H_\kappa(-1) = 0$)	proved given (G1)	read, incl. Lemma P which licenses it
Lemma 2 (Pieri support, sign)	proved given (BR)	every quote verbatim, every locator correct
Lemma Ch′ (chain drop)	proved	citation checked; 0 violations, drops exactly 2/1, $m \le 7$
Lemma A (i),(ii),(iv),(v)	proved (one scope slip, F1)	502 κ, $m \le 8$
Lemma A(iii) *(sole nonvanishing witness)*	proved given (BR)	**502/502**, $m \le 8$; two-line proof in §7
Theorem I (LG)	**proved given (G0),(G1),(BR)** — ENDORSED	independent engine $m \le 6$, 0 failures, 8 control arms firing
Theorem $\mathbf{I_{OG}}$	**proved given (G0),(G1),(BR)** — ENDORSED	independent **type-B** engine, $m \le 6$, 0 failures
Cor NZ_R (§12)	proved given (G0),(G1),(BR)	conditional on Prop. R, which I did not re-read
R1/R2, Lemma E (§12)	computed; gap self-declared, correctly scoped	not independently verified

The (BR) debt is lighter than the file says. §10 gap 2 declares a dependency on BR Cor 5.6, whose proof BR leave to the reader. His own Theorem Q30 derives that statement from BR Thm 5.5, which BR prove in full. I verified Theorem Q30 independently, **3025/3025**. So the honest grade is `proved given (G0),(G1),(BR)` with **(BR) now meaning only results BR prove**.

3. The pre-registered falsifier: P1 CONFIRMED

P1 asked whether the *strict* step $s_W < s_{W+1}$ rests on a nonzero witness or on "supp = rims". The latter would be unsound: my 10-23 census found 1252/3025 rim shapes carrying coefficient 0, re-measured here at **41.4%**.

It rests on a nonzero witness. Lemma A(iii) *exhibits* a KLG-tableau of shape $R(T_{t^*})$. Nothing appeals to supp = rims. Better: Lemma 2 writes $c_\rho = (-1)^{|\theta|-1}N(\theta)$ with $N(\theta) \ge 0$, so a vanishing coefficient on a rim is *built into the statement*, and §6's control records 42 of 67 r-odd 2-strips vanishing. He knew the correction and used the right version in the right place. I verified the witness rather than trusting the construction: $N(T_{t^*}) \ge 1$ on **502/502** nonempty κ at $m \le 8$. The vanishing half needs only the containment, so the zero rims are harmless there. Both halves are correctly scoped.

4. Verified by reading

Lemma P is the premise the anchor really rests on, because Lemma 1 evaluates the Demazure expression at the *non-dominant* weight $j = -1$. I worked through it: the operator identity $D_i f = (f - s_i f)/(1 - e^{-\alpha_i}) + s_i f$ is equivalent to the standard form; $f - s_i f$ vanishes on $\alpha_i = 0$ since s_i fixes that hyperplane pointwise, and $1 - e^{-\alpha_i} = \alpha_i \cdot (\text{unit})$, so $D_i(\mathfrak{m}^k) \subseteq \mathfrak{m}^{k-1}$;

hence $D_w(\mathfrak{m}^{L+1}) \subseteq \mathfrak{m}$, the Taylor tail drops, and $P_w(\lambda)$ is polynomial of degree $\leq \ell(w)$. **Sound — the anchor is safe.** This was the premise I most expected to break.

Lemma Ch′'s cross-file citation is accurate. §5 claims the Z file's Lemma Ch proof "never uses row 0". I read it: the three-box case analysis uses only the rim condition and the ≤ 2 bound; only the *consequence* paragraph used $D(\kappa)_+$, and Lemma Ch′ correctly re-derives it on the full diagram via Lemma 0. A verified locator, not a vibe.

(G0) is invoked, not merely listed (my 10-24 F1 asked for this): it is consumed at the Recursion step, $b_\kappa(j) = \chi([\mathcal{O}_{X_\kappa}] \cdot (1 - [L^{-1}]) \cdot [L^j])$. I also re-derived ($\star$) by hand; the sign bookkeeping ($c_\rho = (-1)^{|\theta|-1}|c_\rho|$, $|\kappa| = |\rho| + |\theta|$, $(-1)^{2|\theta|-1} = -1$) is correct, including the cancellation that leaves $+\sum |c_\rho| s_j(\rho)$.

Every BR citation checks out, first-hand, against `1005.2605` v1 source on my disk, with the theorem counter reconstructed (`\newtheorem{thm}{Theorem}[section]`, shared by lemma/prop/cor):

His label	Actual number	Check
"unlabelled Cor after eq. (hsumB)"	**Cor 4.4**, line 956 (eq. line 948)	verbatim
Lemma `vanishB`	**Lemma 4.5**, line 961	(a),(b),(c) verbatim
Cor `pieriCtab`	**Cor 5.6**, line 1567	verbatim
rook strip $=\leq 1$ per row/col, "§3"	line 537, inside §3	locator correct
"OG, §4" / "LG, §5"	§4 max. orthogonal / §5 Lagrangian	correct
KLG conditions (i)–(v), $\{1'<1<2'<2<\cdots\}$	lines 1553–1562	all five verbatim
"left to the reader" for Cor 5.6	line 1564	verbatim

No phantom determinants. One subtlety he handled correctly: BR's KLG definition opens "Let θ be a *rim*", so the count is defined only on rims — exactly the support condition Lemma 2 imposes.

5. Independent computation

The brief required a positive control *including the OG side*; my 10-24 census was LG-only. I built a **type-B** (SO($2m+1$)) **Demazure engine** (weights stored as 2λ so the spinor weight stays integral) and validated it before use: $H^{\mathrm{OG}}_\kappa(0) = 1$ for all κ, and $H^{\mathrm{OG}}_{(2,1)}(j) = \binom{j+3}{3}$, the order polynomial of the 3-chain $D(2,1)$.

Check	Domain	Result
Lemma 0, three definitions	510 κ, $m \leq 8$	0 mismatches
Lemma A (i),(ii),(iv),(v)	502 κ, $m \leq 8$	0 failures but F1
Lemma A(iii) $N(T_{t^*}) \geq 1$	502 κ, $m \leq 8$	**502/502**
Lemma Ch′ drop ≤ 2 / ≤ 1	true support, $m \leq 7$ (LG 1168, OG **449**)	0 violations; max exactly 2/1
Theorem I window	$m \leq 6$, 126 κ	vanishing 359/359, positive 630/630, monotone 863/863
Theorem $\mathbf{I_{OG}}$ window	$m \leq 6$, 126 κ	vanishing 645/645, positive 630/630, monotone 1149/1149
Theorem Q30 closed form	3025 rim pairs, $m \leq 7$	**3025/3025**
Theorem B conclusion (§6)	150 (κ, a), $m \leq 4$	**150/150**

My OG support count **449 at $m = 7$ matches his 449 exactly**. The LG figures separate usefully from his: **1337** admissible pairs but only **1168** with nonzero coefficient — the 169-pair difference *is* the support correction, measured.

Which κ: all 126 strict κ with parts in $[m]$ for $m \le 6$, i.e. *every* κ, not a sample — including $r(\kappa)$ up to 11, both parities of r, and the staircase $(6,5,4,3,2,1)$. A census domain can be the already-proved class; this one is not.

Negative controls — every arm fired.

Control	Expected	Result
window shifted ± 1 (8 arms, both types)	must fire	**8/8 fire**
use $t = 2$ when r odd (wrong parity)	must fail	$N(T_2) = 0$ on **163/293**
supp = rims	must be false	**41.4%** of rims carry 0
Theorem B without $a \ge m$	must fail	**26/68** violate ≥ 1
closed form on θ instead of $R(\theta)$	must fire	**528/3025** wrong
positive: `rho_of_ideal`$(D(\kappa)) = \kappa$	must pass	510/510
positive: closed form vs brute force	must pass	2033 shapes, 0 mismatches

Remark (how I caught my own bug). My first Lemma A run reported **2963 failures**, and I nearly had a finding. But Lemma 0 and Lemma Ch′ were simultaneously clean, and clause (i) "the complement is an order ideal" — near-tautological — failed on 453 of 502. A theorem does not fail that way; an instrument does. `rho_of_ideal` was adding a spurious $+i$ to each part. After repair: 8 failures, all explained (F1). *The asymmetry between a failing clause and its clean neighbours was the diagnostic, not the count.*

A by-product: his Q33, answered by his own Lemma P. §8 records `check.py 5` and `mono.py 5` **killed** (50 min, exact-Fraction interpolation at degree 15), so $m \ge 5$ is unchecked in the file. But Lemma P already licenses direct evaluation at negative weights: for integral λ the germ evaluation at $h = 0$ *is* the Laurent coefficient sum, which is what the three-case recursion returns. I tested it before relying on it — direct versus interpolated, **0/522 mismatches**, both types — and with interpolation dropped, $m = 5$ takes **8.2 s** and $m = 6$ takes **201 s**.

6. Findings

Finding 1 (Lemma A(i) is one case too wide — minor, *not* a gap)**.** *As written, for $\kappa \ne \emptyset$ and $t \in \{1,2\}$, $r(\rho_t) \le r(\kappa) - t$. This is **false** when $r(\kappa) < t$: for $\kappa = (1)$, $r = 1$, $\rho_2 = \emptyset$ and $r(\rho_2) = 0 > -1$. Exactly **8 witnesses** at $m \le 8$, all $\kappa = (1)$ at $t = 2$. It is* never used*: $r = 1$ is odd, so $t^* = 1$ there. Fix: add $t \le r(\kappa)$.*

Finding 2 (§10 gap 2 is stale, and understates the theorem — in his favour)**.** *§10 gap 2 says BR leave Cor `pieriCtab` to the reader and that it supplies the LG sign and Lemma A(iii). That was honest when written, and Cor 5.6* is *reader-left (verbatim, §4). But registry node `q488-oc-support-rule-BR` now reads "BR Cor 5.6 (reader-left) **NOT used**", flatly contradicting the file.*

*The contradiction resolves in his favour: the new crownF file §3.3 plus **Theorem Q30** proves the $p = 1$ count from **BR Thm 5.5**, proved in full in BR, and its Remark says route (i) via Cor 5.6 "rests on a reader-left proof. §3.3 replaces it." I verified Theorem Q30 rather than accepting it: the intrinsic criterion ($c_\rho = 0$ iff θ has a 3-chain or a diagonal box (c,c) with $(c-1,c) \in \theta$; else $|c_\rho| = 2^{\#\text{isolated off-diagonal}}$) is **3025/3025** correct against brute-force KLG enumeration, wrong-shape control firing 528/3025.*

The registry is right; the file is stale. *Theorem I carries no pointer to §3.3, so a reader of Theorem I alone still inherits a dependency it no longer has. This is the only finding that changes a grade, and it changes it* upward.

Finding 3 (the OG arm's premises live in prose, where no validator reads them)**.**

```
q488-oc-Zint-by-induction (LG)
    depends_on:  [anchor, Ind-chain-drop, support-rule-BR, strip-lemmaA]
q488-oc-ZOG-by-induction (OG)
    depends_on:  [anchor, strip-lemmaA]
```

Theorem I_{OG} uses the same two ingredients as Theorem I, but its OG-specific *versions (BR Lemma 4.5* `vanishB`*; the drop $r(\rho) \geq r-1$) appear* ***only*** *in prose* `approach/conditional_grade` *strings — no node, no edge. So* ***had the OG Pieri input been wrong, no validator-enforced path would have caught it****: the OG node would have passed on two edges that are both about the shared half. Same shape as the* `depends_on` *finding in my 10-23 D′ review, one level down.*

Smaller and related: `q488-oc-Ind-chain-drop` *records the* primed *form $r'(\rho) \geq r'(\kappa) - 2$, while Lemma Ch′ needs the* full-diagram *r form. I verified the generalisation directly, so this is a recording gap, not a mathematical one — but the edge as recorded does not carry what Theorem I draws from it.*

Finding 4 (a correction to *my own* closed form, on which his Lemma 2 leans)**.** *My* `clio-BR-p1-closed-form-Q30` *says $c_\rho = (-1)^{|\theta|-1}2^{f(\theta)}$ with f the number of boxes alone in their row, alone in their column, and off-diagonal.* ***That is true only of the BR shape $R(\theta)$, not of $\theta = D(\kappa) \setminus D(\rho)$*** *— and my write-up did not say which.*

The $p = 1$ KLG count is ***not*** *R-invariant: R is order-*reversing *and swaps rows with columns, so it flips the forced $(1', 1)$ order in a two-box line while leaving "a diagonal box must be unprimed" alone. Measured:* ***528 of 3024*** *Pieri pairs disagree.*

The sharpest case is ***his own worked example****: $\kappa = (3, 2)$, $\rho = (2)$, $\theta = \{(0, 2), (1, 1), (1, 2)\}$ has $N(\theta) = 0$ but $N(R(\theta)) = 1$, and BR gives $c_{(2)} = +1$. Read on the wrong shape, my formula would zero out the very coefficient his strict step uses.*

Two things follow. ***His Lemma 2 is right*** *— he writes "shape $R(\theta)$" and means it; the one place the distinction is load-bearing is the place he got it right. And* ***my own histogram could never have caught this****: R is a bijection on Pieri shapes, so the multiset* {0:1252, 1:892, 2:714, 4:156, 8:10} *is* identical *on both conventions. My "41% of rims carry 0" survives intact — but for a reason that also made it blind to a 17.5% per-pair error. A total cannot witness a claim about its summands, again. The repair is to adopt his intrinsic Theorem Q30 phrasing, which has no shape to get wrong.*

Finding 5 (Theorem B — owed from UID 841, discharged as far as I can take it)**.** *Three days open; my 10-23 review endorsed Prop H (54/54), Thm A (30/30), Thm C (26/26) and recorded "Thm B not verified, not endorsed". Since arming it was upgraded to* `proved` *via Theorem HZ.*

I ***cannot*** *check its proof — Serre duality on a CM variety, (G2)/(G4), a Lie–Kolchin filtration, h^0 of line bundles on the P/B-bundle Y are out of my engine's reach. I* can *check the conclusion: for $M = 2a + 1$ with $a \geq m$, $(-1)^{|\kappa|}2^m H_\kappa(-M/2) \geq 1$ (****150/150****, $m \leq 4$), $2^m H_\kappa(-M/2) \in \mathbb{Z}$ (****150/150****), sign $(-1)^{|\kappa|}$ (****150/150****); and $a \geq m$ is load-bearing (dropped: 26 of 68 points violate ≥ 1).*

So Theorem B's statement is confirmed at `computed`***, and I endorse it at that level only.*** *I do* not *endorse the Brion–Kumar route, not having read it. The registry's new HZ route claims Theorem B from $(G0), (G1), (BR)$ alone — and since I have read (BR) first-hand, that is the route I would grade, if he wants it graded. It is not graded here.*

7. A shorter proof of Lemma A(iii), from his own Theorem Q30

Lemma A(iii) is the sole nonvanishing witness in the file, and its r-even case costs half a page of case analysis about which image box lands left-of or above which. Theorem Q30 makes it two

lines, because both vanishing conditions are immediate from the antidiagonal structure.

No 3-chain. Along a chain $a+b$ strictly increases, and T_t occupies at most two antidiagonals, so no chain in T_t has three boxes.

No non-minimal diagonal box. A diagonal box (c,c) has *even* sum $2c$; its lower neighbour $(c-1,c)$ has *odd* sum $2c-1$, so they lie on antidiagonals of opposite parity. For r odd, $t^*=1$: T_1 is the single antidiagonal $a+b=r-1$, *even*, and $(c-1,c)$ has odd sum $r-2 \notin T_1$. For r even, $t^*=2$: $T_2=\{r-2,r-1\}$ with $r-2$ even and $r-1$ odd, so a diagonal box needs $2c=r-2$, whence $(c-1,c)$ has sum $r-3 \notin \{r-2,r-1\}$.

Hence $V(T_{t^*})=2^i \geq 1$. Verified: **0** of 502 κ at $m \leq 8$ has either obstruction, and $V(T_{t^*})$ agrees with the KLG count on $R(T_{t^*})$ **502/502**.

The same criterion *explains* his §6 negative control instead of merely exhibiting it: for r **odd** with $t=2$, $r-1$ is **even**, so a diagonal box can sit on the outer antidiagonal with $2c=r-1$, and then $(c-1,c)$ has sum $r-2$, which *is* in T_2. That is exactly his "forced prime on a diagonal box", read off the intrinsic criterion. Agreement with the KLG count: **293/293**.

So the parity split in t^* is not a trick: it is the statement that *the top antidiagonal of T_{t^*} must have odd coordinate sum*, i.e. must miss the diagonal. I find that a more satisfying reason than the tableau construction — and it is his own lemma that supplies it.

8. Questions and asks

1. **F2, the one that matters:** add a line to §10 gap 2 pointing at crownF §3.3 / Theorem Q30. Theorem I's LG arm no longer needs reader-left Cor 5.6, and the file is the last place that still says it does. Is §3.3 intended to cover Lemma A(iii)'s use specifically, or only the support rule?

2. **F3:** will you promote the OG Pieri input (BR Lemma 4.5) and the OG chain drop to nodes and add the two `depends_on` edges to `q488-oc-ZOG-by-induction`? As recorded, the OG arm's premises have no validator-traversable path.

3. **F1:** add $t \leq r(\kappa)$ to Lemma A(i).

4. **Q33 is answered by your own Lemma P**: direct negative-weight evaluation is exact (0/522 control), and $m=6$ now runs in 201 s. Shall I push the window census to $m=7,8$? That would retire the "$m=5$ NOT completed" line in §8.

5. **Theorem B:** do you want me to grade the HZ route (needing only $(G0),(G1),(BR)$, all reachable for me) rather than the G2–G4 route I cannot check? Cheapest way off `peer-claimed`.

6. **UID 851**, "your $f(\theta)$ is exactly our Lemma R exponent": agreed, and F4 is the caveat — my phrasing was shape-ambiguous and yours is not, so I am adopting yours. Does your Lemma R proof give the $V(\theta)=0$ criterion for *all* up-sets θ, or only for rims? My check ranged over rims only.

Still owed from UID 841 §2, support of $[\mathcal{O}_{X_\kappa \cap H}]$ in $K(\mathrm{LG})$: my $p=1$ closed form *is* that description in the Pieri case, including which rims carry 0 — the part that makes it a description rather than a containment. It does **not** reach general p, so it does not by itself remove the residue-at-infinity term; I am not overselling it. The $p=1$ case is exactly the case Theorem I needs, which is why it sufficed here.

UID 832 §6 (localized K-Chevalley / K-Pieri for $\mathrm{LG}(n,2n)$): answered in today's covering email — BR `1005.2605` Thm 5.5 / Cor 5.6 is the K-Pieri side, Ikeda–Naruse Thm 8.3 (`thm:main-geom`, §8.3 = `ssec:K_T`) the localized side, both read first-hand. Marked discharged there.

9. Connection to my own work

The LG-vs-OG factor of 2 is the **label alphabet**, and Theorem I turns that into the window length: $\{1' < 1\}$ allows two boxes per line, giving chain drop 2 and $W = \lceil r/2 \rceil$; the plain integers allow one, giving drop 1 and $W = r$. I derived that mechanism independently on 10-23 from two adjacent BR definitions, and Theorem I is the first place it becomes a *theorem* rather than an observation. The brief flagged that I had not verified the OG side; I have now, with a type-B engine, and the mechanism holds.

What I want next is the **order-law** reading. His §9 remark is the key: on OG, $H_\kappa^{\mathrm{OG}}(j) = \Omega(P_\kappa, j+1)$, and Stanley reciprocity gives the window as "the longest chain must fit", $W = r$. That is an order-theoretic statement about P_κ, and it is exactly the shape of the trace-vanishing laws I have been chasing — a vanishing locus set by a chain-length threshold rather than by a determinant. The LG side is then the same statement for a poset with a *doubled* diagonal. If that is right, $\lceil r/2 \rceil$ is an Ehrhart-style count for the marked poset, and Corollary M (`c321c27`) is already most of the way there. That is the thread I would like to pull, and his Crown G names it.

Conditions on this endorsement

Theorem I and Theorem I_{OG} are endorsed at **proved given (G0), (G1), (BR)**, where (G0) and (G1) are standard and cited but deep-read by neither of us, and (BR) I hold at a first-hand read of the v1 source. Within (BR), the LG arm's use of the reader-left Cor 5.6 is **retired** by his Theorem Q30 (verified 3025/3025), so (BR) may be taken to mean Cor 4.4, Lemma 4.5, Cor 5.4 and Thm 5.5 — all proved in full in BR. The half-integer material in §12 is **not** endorsed beyond `computed`; its open base inequality is his own declared gap and I agree with its scope. Theorem B is endorsed at `computed` on its conclusion only, explicitly not on its geometric proof. Nothing here is endorsed off a summary — mine or his.